Exercise Sheet 11

Throughout, $\mathfrak{g}$ is a finite-dimensional complex Lie algebra with ordered basis $(x_1, \dots, x_n)$, and U_p is the filtration of $U(\mathfrak{g})$ from Lecture 11. The algebras $U(\mathfrak{g})$ and their modules may be infinite-dimensional. In $U(\mathfrak{sl}_2)$ use the ordered basis (f, h, e) and the rewriting rules

$$eh = he - 2e, \qquad hf = fh - 2f, \qquad ef = fe + h.$$

The exercises combine calculations in $U(\mathfrak{g})$ with proofs of the filtration, duality and PBW properties.

Exercise 11.1 (Filtrations).

(i) Without using PBW, prove that $ab - ba \in U_{p+q-1}$ for $a \in U_p$ and $b \in U_q$.

(ii) Deduce that $\operatorname{gr} U(\mathfrak{g})$ is commutative and that the map $\psi\colon \operatorname{Sym}(\mathfrak{g}) \to \operatorname{gr} U(\mathfrak{g})$ of the lecture is surjective.

(iii) Prove that ψ is injective if and only if the ordered monomials are linearly independent in $U(\mathfrak{g})$.

(iv) Using PBW, compute $\dim U_p(\mathfrak{sl}_2)$ for $p = 0, 1, 2, 3$.

Exercise 11.2 (Powers in $U(\mathfrak{sl}_2)$).

(i) Prove $e^j h = (h - 2j)e^j$ and $hf^j = f^j(h - 2j)$ for $j \geq 0$.

(ii) Prove, for $k \geq 1$,

$$e^k f = fe^k + khe^{k-1} - k(k-1)e^{k-1}.$$

(iii) Write $e^2 f$ in the basis $f^a h^b e^c$. Give the class of $e^2 f$ in U_3/U_2 and the class of $e^2 f - fe^2$ in U_2/U_1.

Exercise 11.3 (Antipode and duals).

(i) Prove that $S^2 = \mathrm{id}_{U(\mathfrak{g})}$.

(ii) In $U(\mathfrak{sl}_2)$, compute $S(ehf)$ directly. Then apply S to the expansion $ehf = fhe - 4fe + h^2 - 2h$ and check that the two answers agree.

(iii) Let V be a $U(\mathfrak{g})$-module. Prove that $(u \cdot \varphi)(v) = \varphi(S(u)v)$ defines a left $U(\mathfrak{g})$-module structure on V^*.

(iv) Let ρ be the defining representation of $\mathfrak{sl}_2$ on $\mathbb{C}^2$, and let $\tau(x)\varphi = \varphi \circ \rho(x)$ on $(\mathbb{C}^2)^*$. Show that τ is not a representation.

Exercise 11.4 (Primitive elements). An element $u \in U(\mathfrak{g})$ is *primitive* if $\Delta(u) = u \otimes 1 + 1 \otimes u$. Let P be the set of primitive elements.

(i) Prove that P contains $\mathfrak{g}$ and is closed under $[u, v] = uv - vu$.

(ii) Let $\mathfrak{g} = \mathbb{C}x$ be one-dimensional. Prove that $U(\mathfrak{g}) = \mathbb{C}[x]$ and $P = \mathbb{C}x$.

(iii) For general $\mathfrak{g}$ and $0 \neq x \in \mathfrak{g}$, prove $x^2 \notin P$. Where is PBW independence used?

Exercise 11.5 (Why the Jacobi identity is needed). Let V have basis x, y, z and define an antisymmetric bilinear bracket by

$$[x, y] = x, \qquad [y, z] = y, \qquad [z, x] = z.$$

Let $A = T(V)/I$, where I is generated by all $a \otimes b - b \otimes a - [a, b]$, and let $j \colon V \to A$ be the natural map.

(i) Compute the Jacobi sum $J = [x, [y, z]] + [y, [z, x]] + [z, [x, y]]$.

(ii) Prove that $j(J) = 0$.

(iii) Prove that $j = 0$ and $A \cong \mathbb{C}$.

(iv) Explain why the construction of the polynomial module in the proof of PBW must fail for this bracket, and at which step.

Exercise 11.6 (A triangular factorisation for $\mathfrak{sl}_2$). Let $\mathfrak{b} = \operatorname{span}\{h, e\}$ and $\mathfrak{n}_- = \mathbb{C}f$.

(i) Prove that every $u \in U(\mathfrak{sl}_2)$ has a unique expression $u = \sum_{a} f^a b_a$ with $b_a \in U(\mathfrak{b})$, almost all zero.

(ii) Prove that $U(\mathfrak{sl}_2)e \cap \operatorname{span}\{f^a : a \geq 0\} = 0$.

(iii) Show $ef^2 = f^2e + 2fh - 2f$. For $c \in \mathbb{C}$, deduce

$$ef^2 - 2(c-1)f \in U(\mathfrak{sl}_2)e + U(\mathfrak{sl}_2)(h - c).$$